CIS 5500: Database and Information Systems
Homework 4b: Relational DB Design

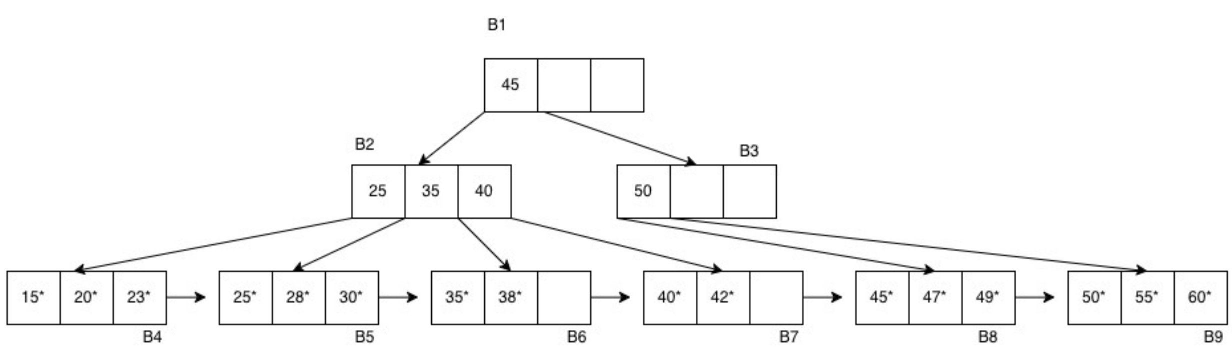
April 6, 2026

Mustafa Rashid

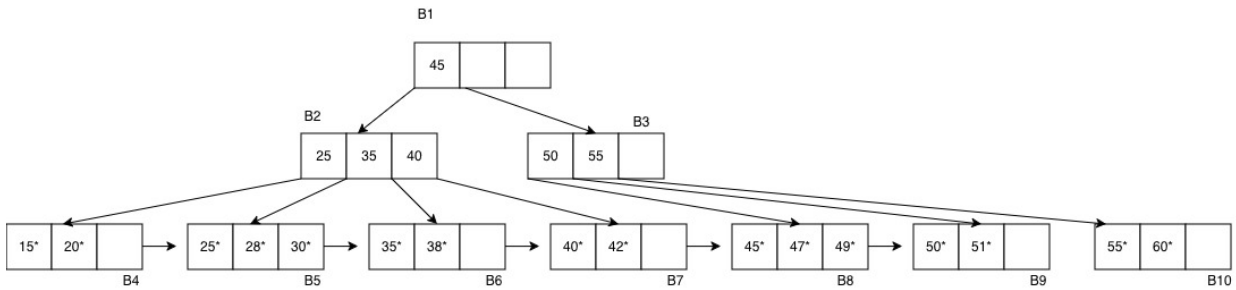
Spring 2026

A. **(14 points)**

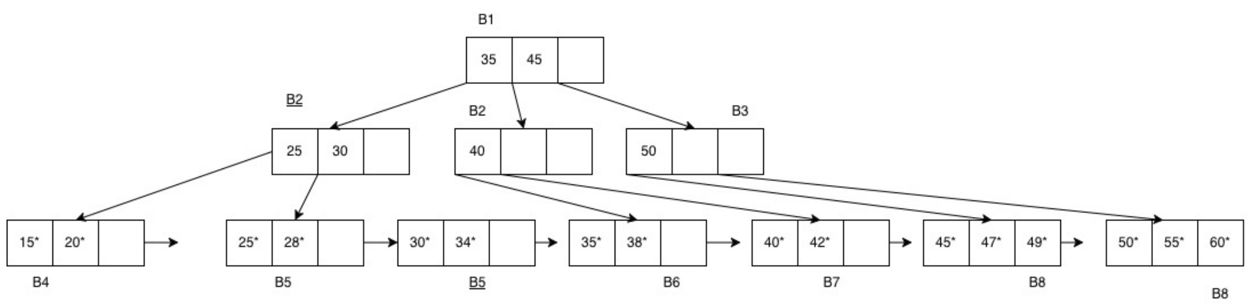
1. Start at B1. Since $39 < 45$, go left to B2. In B2, since $35 \leq 39 < 40$, follow the pointer between 35 and 40 to B6. B6 contains 35^* and 38^* but not 39^*, so the record is not found. Blocks read: B1, B2, B6.
2. Start at B1. Since $32 < 45$, go left to B2. In B2, since $25 \leq 32 < 35$, follow the pointer between 25 and 35 to B5. B5 contains 25^*, 28^*, 30^*, none of which satisfy ≥ 32. Scan right to B6, which contains 35^* and 38^* (both qualify). Continue right to B7, which contains 40^* and 42^* (both qualify). Stop since the next value (45^*) is not strictly less than 45. Blocks read: B1, B2, B5, B6, B7. Qualifying entries: 35^*, 38^*, 40^*, 42^*.
3. Navigate B1 $\rightarrow$ B2 $\rightarrow$ B4 (since $23 < 25$, follow the leftmost pointer in B2). B4 $= [15^*, 20^*]$ has 2 entries (below the maximum of 3), so 23^* is inserted directly without splitting. No structural changes occur; B4 becomes $[15^*, 20^*, 23^*]$.



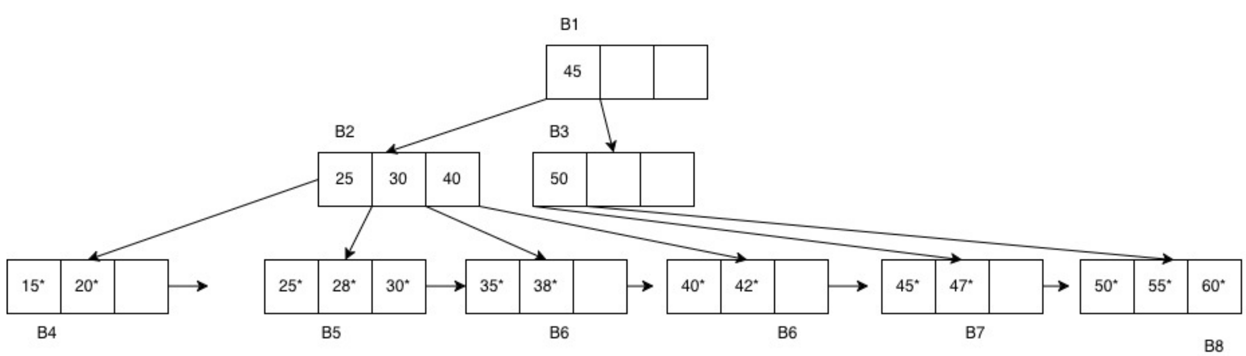
4. Navigate B1 $\rightarrow$ B3 $\rightarrow$ B9 (since $51 \geq 45$ and $51 \geq 50$). B9 $= [50^*, 55^*, 60^*]$ is full (3 entries). Inserting 51^* gives $[50^*, 51^*, 55^*, 60^*]$ (overflow). **Split B9**: left B9 $\leftarrow [50^*, 51^*]$, new block B10 $\leftarrow [55^*, 60^*]$; push key 55 up to B3. B3 $= [50]$ becomes $[50, 55]$ with children B8, B9, B10 (2 keys; no overflow).



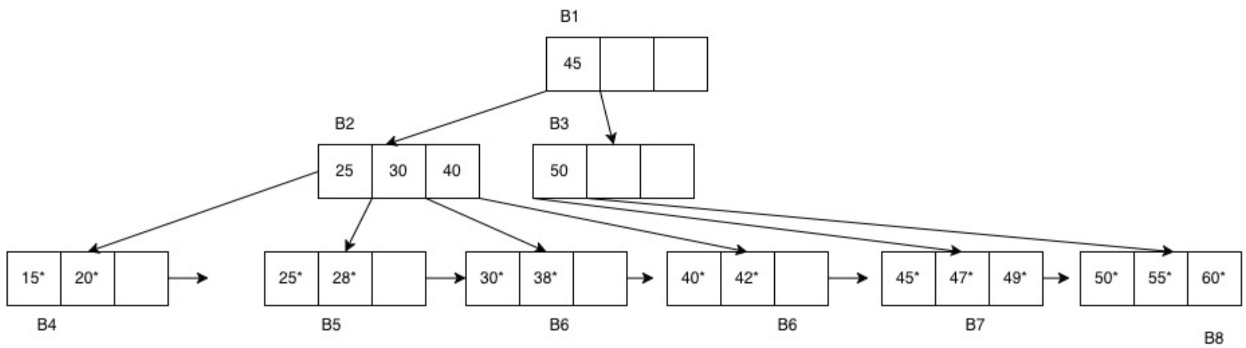
5. Navigate B1 → B2 → B5 (since $25 \leq 34 < 35$). B5 = $[25^*, 28^*, 30^*]$ is full. Inserting 34^* gives $[25^*, 28^*, 30^*, 34^*]$ (overflow). **Split B5**: left B5 ← $[25^*, 28^*]$, new B10 ← $[30^*, 34^*]$; push 30 up to B2. B2 = $[25, 35, 40]$ becomes $[25, 30, 35, 40]$ (overflow). **Split B2**: push 35 up to B1; left B2 ← $[25, 30]$ with children B4, B5, B10; new B11 ← $[40]$ with children B6, B7. B1 = $[45]$ becomes $[35, 45]$ with children B2, B11, B3 (no overflow).



6. Navigate B1 → B3 → B8 (since $49 \geq 45$ and $49 < 50$). B8 = $[45^*, 47^*, 49^*]$ has 3 entries; removing 49^* leaves B8 = $[45^*, 47^*]$ (2 entries = min; no underflow). No structural changes.



7. Navigate B1 → B2 → B6 (since $35 \leq 35 < 40$). B6 = $[35^*, 38^*]$ is at minimum (2 entries); removing 35^* leaves B6 = $[38^*]$ (underflow). Right sibling B7 = $[40^*, 42^*]$ is at minimum and cannot lend. Left sibling B5 = $[25^*, 28^*, 30^*]$ has 3 entries and can lend. **Borrow from B5**: move 30^* to B6. B5 ← $[25^*, 28^*]$, B6 ← $[30^*, 38^*]$. Update separator between B5 and B6 in B2 from 35 to 30: B2 = $[25, 30, 40]$ (no underflow).



B. **(6 points) Answer:** 12

Because all inserted values are ≥ 45, every insertion goes into the rightmost subtree. To increase the height, we must overflow the root, which requires adding 3 new keys to it.

Consider node $B3$, which initially has 1 key (2 children). To make it split and push a key up to the root, it must reach 4 keys, i.e., gain 3 new keys. Its two children ($B8$ and $B9$) are full, so the first two leaf splits take 1 insertion each. After these splits, new leaves have only 2 entries, so the next split requires 2 insertions.

Thus, producing one key for the root costs

$$1 + 1 + 2 = 4$$

insertions.

Since the root needs 3 additional keys to overflow, the total number of insertions required is

$$3 \times 4 = 12.$$